COURSE DESCRIPTION			
Program	**Polymer Technology**		
Course Title	**Polymer Science Lab**		
Course Code	**2099**	**Semester**	**II**
Type of Course	**Lab**	**Contact Hours**	**45**
Teaching Scheme (L: T:P:R)	**0:0:3:0**	**Credits**	**1.5**
CIE Marks	**60**	**ESE Marks**	**40**

Introduction

The Polymer Science lab has been designed to enhance students' understanding of polymers through hands-on experimental learning. The course emphasizes the development of practical skills in the standardization of solutions. Learners will be able to carry out basic polymer synthesis using standard laboratory techniques. The course also explains methods for determining the specific gravity of polymers and helps students interpret the results to understand polymer density.

Course Objectives

1	Determine the specific gravity of polymers using standard experimental methods and interpret the results.
2	Perform basic polymer synthesis using standard laboratory techniques.
3	Record, analyze, and interpret experimental data accurately and systematically.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Synthesis of Polymers, Specific gravity of Polymers	1091	Introduction to Polymer Science	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Prepare standard solutions accurately using an analytical balance.	Apply
CO2	Produce polymers by different polymerization techniques.	Apply
CO3	Determine the specific gravity of plastics and elastomers.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	2	-	-	-	-	-	-	3
CO2	3	-	-	2	-	-	-	-	-	-	3
CO3	3	-	-	2	-	-	-	-	-	-	3
CO4	3	-	-	2	-	-	-	-	-	-	3
Total	12	-	--	8	-	-	-	-	-	-	12
Correlation Level	3	-	-	2	-	-	-	-	-	--	3
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Analytical Balance	Familiarization and study of analytical balance	CO1	Apply	3
2	Standard solution preparation	Prepare standard Oxalic acid solution at different normalities - 0.05 N, 0.1N, 1N.	CO1	Apply	3
3	Standard solution preparation	Prepare standard Sodium carbonate solution at different normalities - 0.1 N, 0.2N, 1N.	CO1	Apply	3
4	Standard solution preparation	Prepare standard Sodium hydroxide solution at different molarities - 0.1 M,1M.	CO1	Apply	3
	Series Exam - I		CO1	Apply	
Module II					
5	Polymerization techniques	Prepare PMMA by bulk polymerization.	CO2	Apply	3
6	Polymerization techniques	Prepare Polystyrene by Suspension polymerization.	CO2	Apply	3
7	Polymerization techniques	Prepare Polystyrene by Emulsion polymerization.	CO2	Apply	3
8	Polymerization techniques	Prepare Urea formaldehyde resin.	CO2	Apply	3
9	Polymerization techniques	Prepare Phenol formaldehyde resin.	CO1	Apply	3
Module III					
10	Specific gravity determination	Find out the specific gravity of polymer sample by Hydrostatic bench method (greater than 1).	CO3	Apply	3
11	Specific gravity determination	Find out the specific gravity of polymer sample by Hydrostatic bench method (less than 1).	CO3	Apply	3
12	Specific gravity determination	Find out the specific gravity of polymer	CO3	Apply	3
	Series Exam - II		CO3	Remember	1.5
	Open ended experiment		CO3	Apply	3

Open-ended experiment

Objective:
To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:
At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:
Modify or extend an existing experiment to explore additional parameters or behaviors.
Integrate multiple concepts from the syllabus into a combined application.
Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Identify common laboratory weighing equipment and understand the importance of calibration and proper taring for accurate measurements.	1.5
2	Understand the importance of standard solutions in titration and quantitative analysis. Read about normality, equivalents, and primary standard characteristics. Understand why oxalic acid is a good primary standard. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
3	Calculate the mass required for preparing the given normality of Sodium carbonate, draw relevant tabular columns. Learn the safety and good Laboratory Practices to be followed while handling chemicals. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
4	Calculate the mass required for preparing the given molarity of Sodium hydroxide,draw relevant tabular columns. Learn the safety and good Laboratory Practices to be followed while handling chemicals. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
Module II		

5	Learn the theoretical concept of bulk polymerization techniques, identify different monomers used for the above process. Study how to remove inhibitors from PMMA. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
6	Learn the theoretical concept of suspension polymerization techniques, identify different monomers used for the above process. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
7	Learn the theoretical concept of emulsion polymerization techniques, identify different monomers used for the above process. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
8	Learn the theoretical concept of condensation polymerization, identify different monomers used for the above process. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
9	Learn the theoretical concept of condensation polymerization, identify different monomers used for the above process. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.	1.5
Module III		
10	Understand the principle of buoyancy in density determination. Identify materials having specific gravity greater than one. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.Relate specific gravity to material properties.	1.5
11	Understand the principle of buoyancy in density determination. Identify materials having specific gravity less than one. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.Relate specific gravity to material properties.	1.5
12	Understand the principle of displacement and its role in density determination. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results.Relate specific gravity to material properties.	1.5
13	Open ended experiment -Identify experiments related to connect with the course outcomes defined and study their theoretical concepts. Complete the laboratory record in a clear and organized way, including all observations, measurements, and results	1.5
14	Study and prepare for lab exams.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Polymer Science	Jayadev Sreedhar, V R Gowariker, N.V Viswanathan	New Age International Private Limited, ISBN-13 : 978-9389802580
2	Practical Chemistry	A. O. Thomas	8th Edition, Scientific Book Centre,Cannanore

REFERENCE			
Sl. No.	Title	Author	Publication
1	Textbook of Polymer Science	Fred W. Billmeyer	Wiley-Interscience; 3rd edition, ISBN-13 :978-0471031963
2	Introduction to Polymers	Robert J. Young, Peter A. Lovell	CRC Press; 3rd edition, ISBN-13 :978-0849339295

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://vlab.amrita.edu/	I
2	https://learncheme.com/screencasts/materials-science/	II
3	https://infinitalab.com/material-testing-service/specific-gravity-of-plastic/	III

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity

1	Physical balance for laboratory use with weight box	Sensitivity 5 mg with 3-stone knife-edge agates.	5
2	Analytical/Digital balances with an accuracy of at least ±0.1 mg	Maximum capacity of 300g with 1mg precision and 0.001g accuracy, Supports multiple units of measurement including grams (g), carats (ct), ounces (oz), and milligrams (mg),Dual LCD display	3
3	Burette 50 mL, Pipette 10mL / Micropipette 10 mL Adjustable, Conical flask (Erlenmeyer) 250 mL, and Retort Stand with Clamp & Boss Head	Borosilicate Glass	6 numbers each (for a batch of 5 students)
4	Magnetic Stirrer With Hot Plate	Capacity : 2 Litre; Speed Range : 200-1500 rpm; Temperature Range : Ambient to 200 °C; Hot Plate : Stainless Steel; Input Voltage : 230V 50Hz; Accessories: Magnetic beads	3
5	Water Bath	Single wall, stainless steel body, 9 holes,10ltr,2kwts without thermostat, Inner dimension- 300×300×100mm	1
6	Hot Air Oven	(125 LTR,3 tray, 1000 Watts, Upto 150°C), micro processor based digital temp controller	1
7	Digital weighing balance with density kit	Accuracy-0.1mg,Capacity-220gm	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to			15	15		
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10

Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student’s ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50